MILITARY SPECIFICATION SHEET

COMPUTER SET, STANDARD AIRBORNE
CP-1539/AYK-14(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

The complete requirements for acquiring the computer described herein shall consist of this specification and the latest issue of MIL-C-85518.

REQUIREMENTS:

System configuration:
SRA location - See Table I.

TABLE I. SRA location in the CP-1539 computer.

SRA	Chassis Location	MIL-C-85518 Reference
DIM	A01	3.6.2.2.1.3
SIM-A	A02	3.6.2.2.1.2
SIM-A	A03	3.6.2.2.1.2
SIM-A	A04	3.6.2.2.1.2
IOP	A05	3.6.2.1.2
PSM	A06	3.6.2.1.1
GPM	A07	3.6.2.1.1
MCM	A08	3.6.2.3.3
CMM-64	A09	3.6.2.3.1.1
CMM-64	A10	3.6.2.3.1.1
PCM-1	A11	3.6.2.4.1
Type 5 Chassis	-	3.6.2.5.1.4
Cooling Adapter	-	3.6.2.5.3

Total weight: 45 lbs. maximum, excluding external cables.

Dimensions: Length 14.00 inches, width 10.125 inches, height 7.625 inches excluding cooling adapter, front panel connectors, and handle. See Figure 50 of MIL-C-85518.

Operating power: 115 Volts ac, 400 Hz (3 phase WYE), 470 Watts/555 VA maximum power requirements shall be based on two of the three MIL-STD-1553 input/output channels operating at 100% duty cycle with alternating 1s and 0s, and the other input/output channel operating in a standby mode.

Reliability in mean-time-between-failure (MTBF): MTBF shall be 620 hours minimum when tested in accordance with 4.4.3.4 of MIL-C-85518.

FSC 7021

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Mean-time-to-repair (MTTR):
Intermediate level SRA replacement: MTTR shall be 20 minutes maximum.

Connector assignments: See Figure 1. Maintenance connector and all unused connectors shall be capped.

Service conditions: As specified in MIL-E-5400 for Class 2X equipment using conditioned air and as modified herein.

Heat dissipation: Electrical heat generated within the unit (Qe) shall be 470 watts maximum. Heat load due to ambient conditions (Qa) shall be 20 watts maximum.

Cooling provisions:
Conditioned air:
Temperature range: -65°F to 120°F with flows specified herein for standard, normal, ground, transit, and emergency cooling as a function of inlet air temperature and maximum computer cooling air discharge temperature of 160°F.
Entrained (free) moisture: Cooling air supplied may contain up to 104 grains per pound of air.

Standard cooling:	0°F,	flow rate = 0.726 lb/min ± 3 percent
	40°F,	flow rate = 0.975 lb/min ± 3 percent
	62°F,	flow rate = 1.185 lb/min ± 3 percent
Normal cooling:	0°F,	flow rate = 0.828 lb/min ± 3 percent
	40°F,	flow rate = 1.160 lb/min ± 3 percent
	62°F,	flow rate = 1.488 lb/min ± 3 percent
Ground cooling:	-65°F,	flow rate = 0.500 to 3.93 lb/min
	0°F,	flow rate = 0.703 to 3.38 lb/min
	62°F,	flow rate = 1.148 to 2.98 lb/min
	100°F,	flow rate = 1.881 to 2.76 lb/min
	120°F,	flow rate = 2.881 minimum lb/min

Transient cooling:
Operating time: 10 minutes ON every 90 minutes with computer cooling air discharge temperature of 160°F.

Cooling air:	0°F,	flow rate = 0.652 lb/min ± 3 percent
	40°F,	flow rate = 0.871 lb/min ± 3 percent
	62°F,	flow rate = 1.063 lb/min ± 3 percent

Before and after: Cooling air shall be standard cooling.

Emergency cooling. Components temperature increased up to rated temperature.

	-65°F,	flow rate = 0.463 to 3.12 lb/min
	0°F,	flow rate = 0.623 to 3.12 lb/min
	40°F,	flow rate = 0.795 to 3.12 lb/min
	62°F,	flow rate = 0.935 to 3.12 lb/min
	100°F,	flow rate = 1.350 to 3.12 lb/min
	147°F,	flow rate = 2.975 to 3.12 lb/min

Minimum emergency cooling air flow rates: See figure 2. Equipment shall sustain no permanent damage or be adversely affected by operation at the percentage of the minimum emergency cooling air flow rates shown for the times specified in Figure 2.

Pressure drop: Forced air cooling circuit pressure drop through the equipment shall be maintained to within a tolerance of ± 15 percent and shall be not greater than the maximum allowance as follows:

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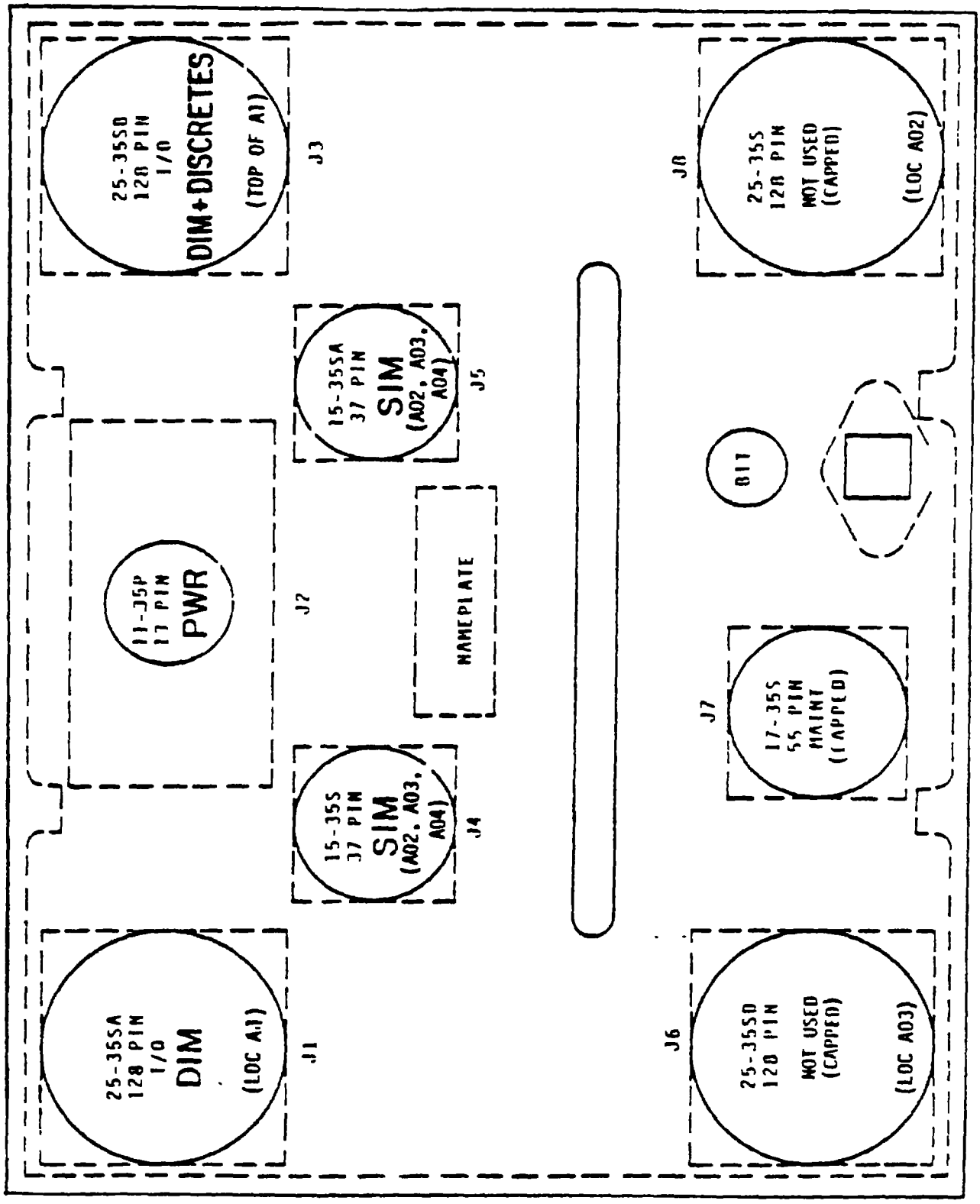


FIGURE 1. CP-1539 front panel.

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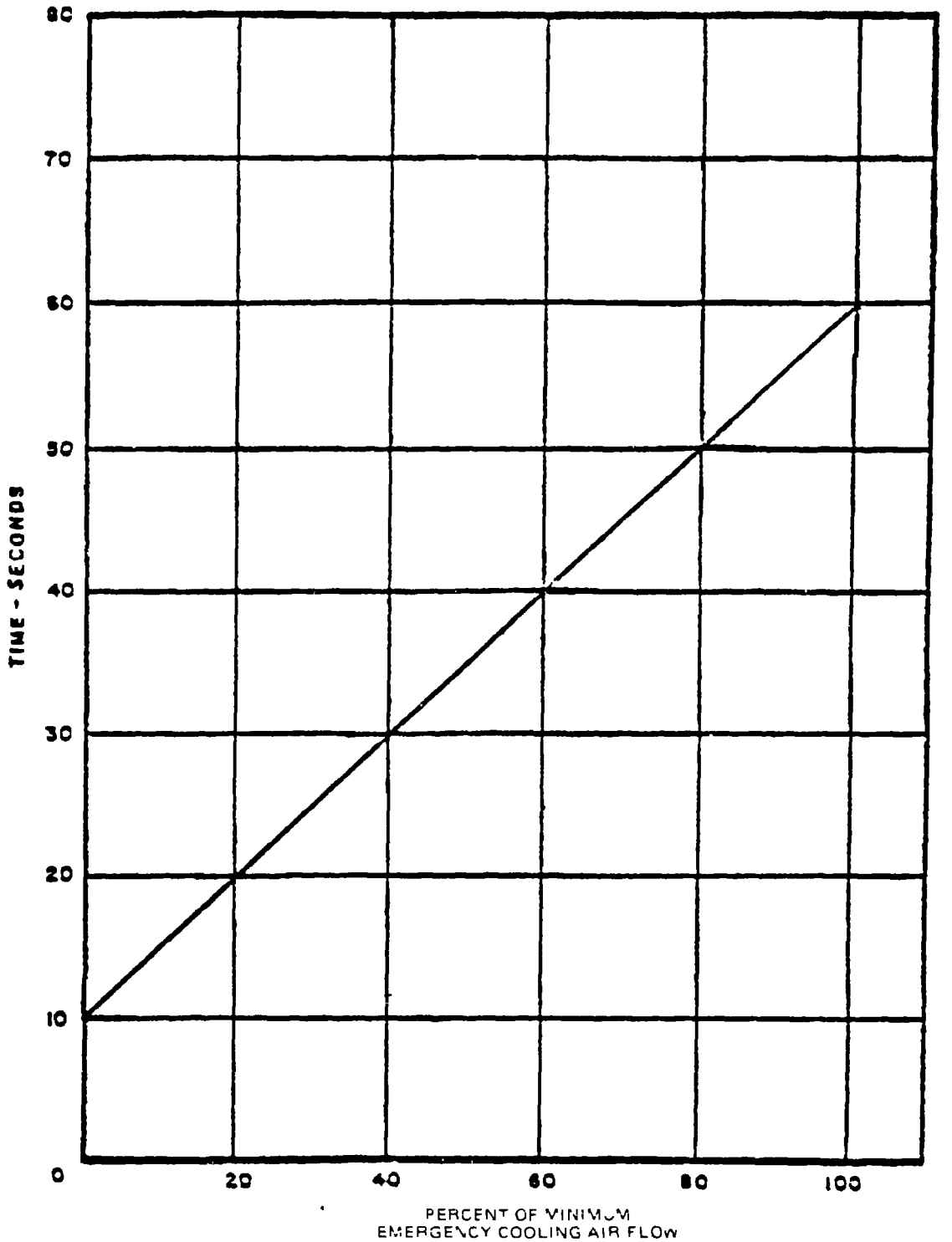


FIGURE 2. Minimum emergency cooling air flow.

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0.60 inches max. of water at 62°F inlet air temperature, 1.08 lb/min and 14.7 psia ambient.

4.0 inches max. of water at 120°F inlet air temperature, 2.65 lb/min and 14.7 psia ambient.

Exterior (touch) temperature: 65°C maximum with ambient temperature of 25°C and normal cooling airflow rates.

I/O channels: The I/O shall be capable of having the DIM and all three SIM channels with active I/O channel programs in execution. The three SIM channels may be independently operated in any modes, except that multiple SIM channels operating as RTs may not receive command words initiating data message sequences within any given 10 microseconds time period. Each bus pair of a MIL-STD-1553 channel shall be assigned to separate connectors designated J4 and J5 in Figure 1. All MIL-STD-1553 input/output bus connectors shall be accomplished via these two connectors. Five chassis ground pins, suitable for terminating the cable shields for MIL-STD-1553 bus connectors, shall be provided on each of the above operational connectors.

Quality assurance provisions. With the exception of AET, when cooling air is required during tests specified in 4.2, 4.3, and 4.4 of MIL-C-85518, the air shall be supplied at a temperature in the range of -65°F to 120°F. The flow rate shall be adjusted to correspond to the selected inlet temperature, an exhaust temperature of 160°F, and the total heat load on the unit.

Preparing Activity
Navy AS

Project 7021-N005-8

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